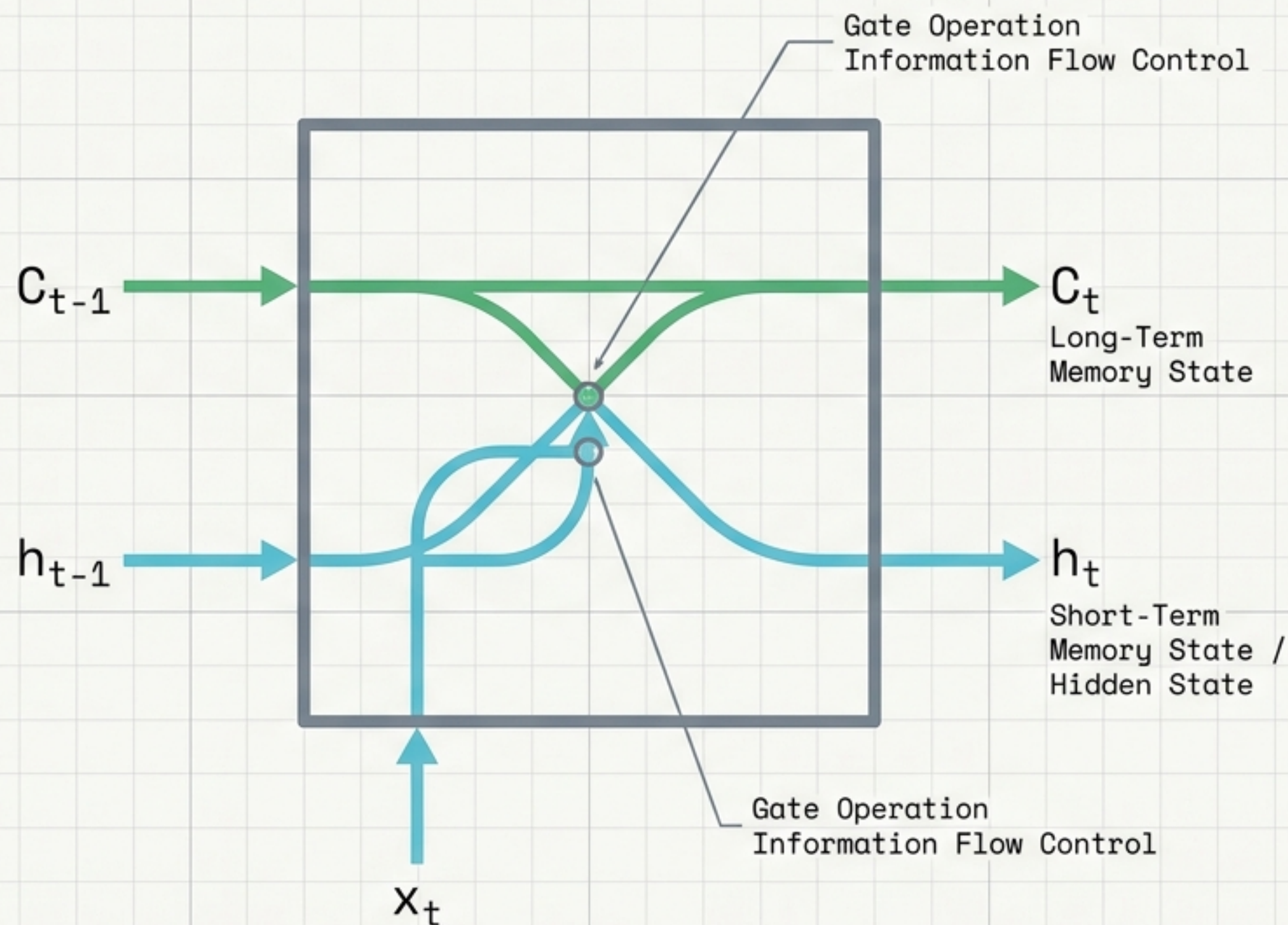


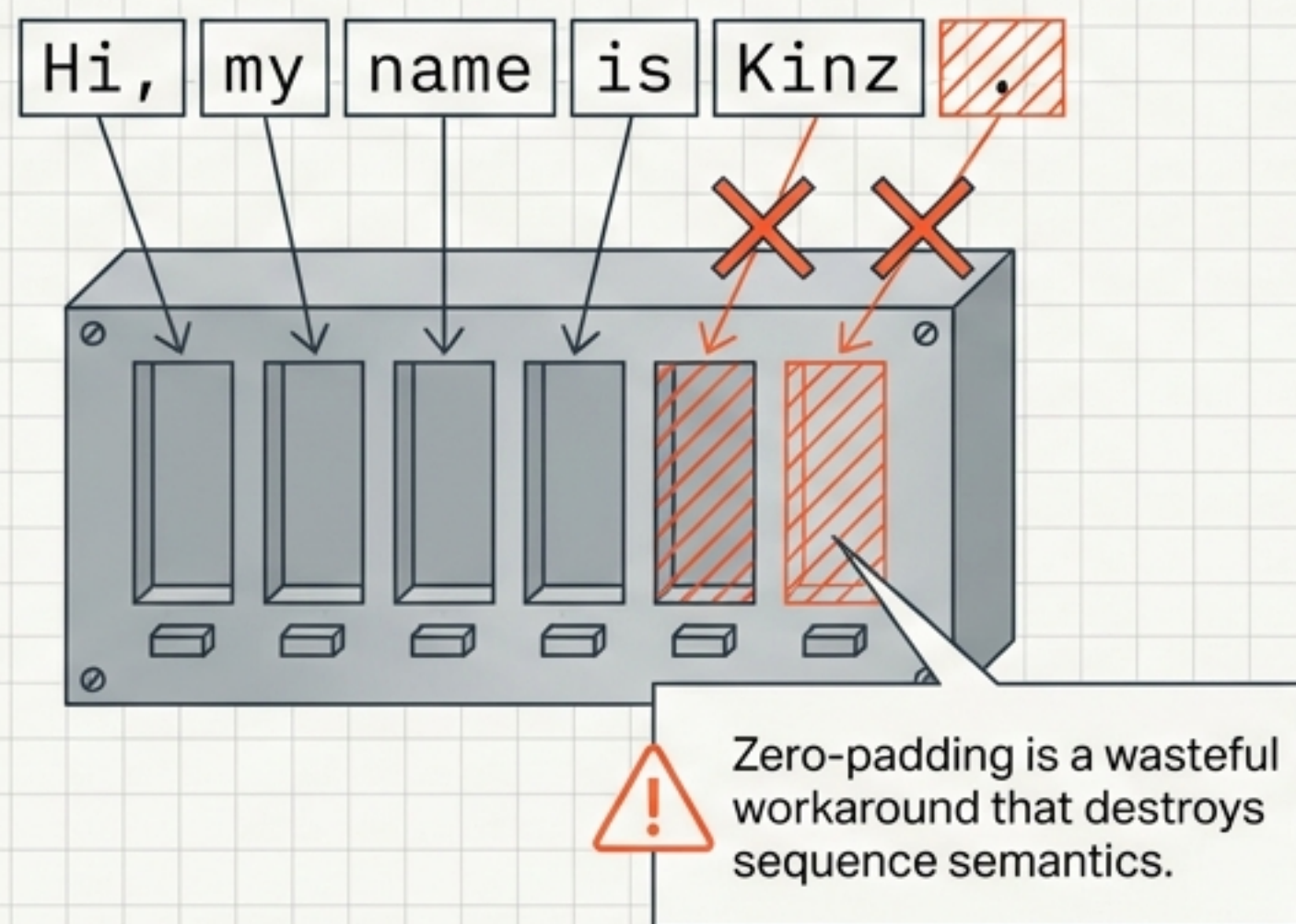
The Blueprint of Memory

Architecting Sequence Models:
From Vanilla RNNs to LSTMs
and Beyond

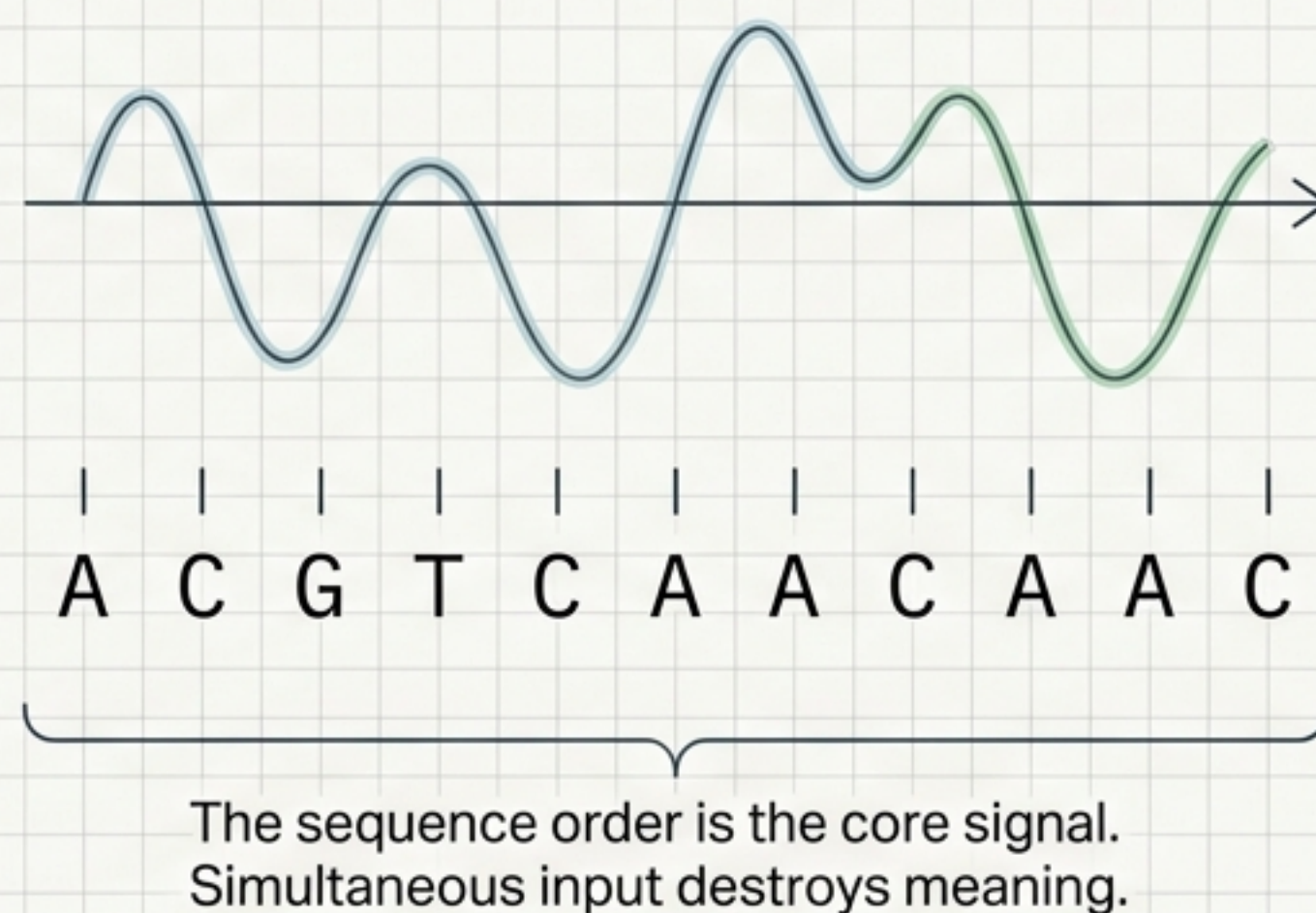


The Structural Failure of Standard Networks

The Rigid ANN



The Dynamic Sequence



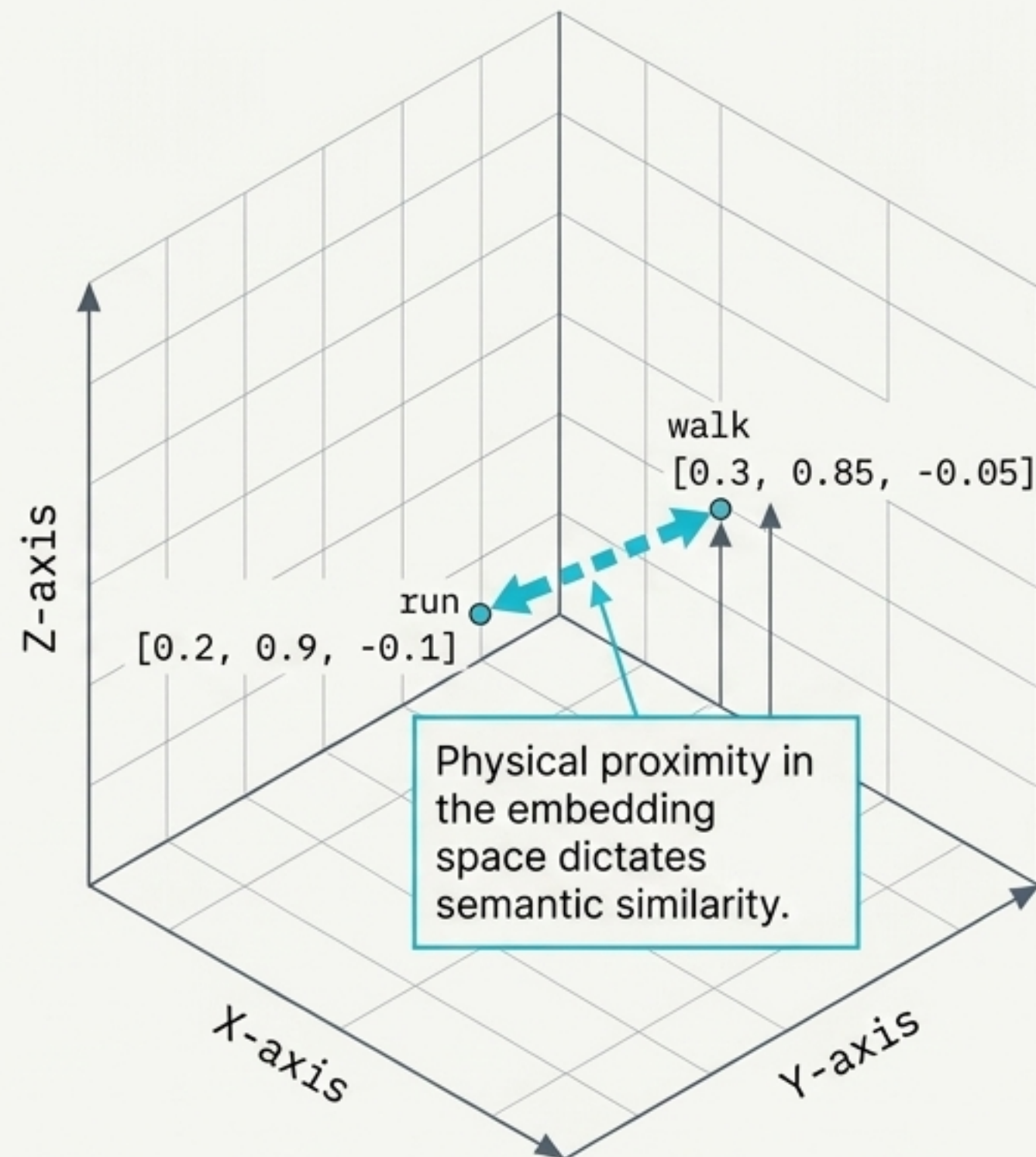
Mapping Words to Mathematics: The Embedding Space

Modern sequence models abandon sparse binary matrices in favour of **Learned Embeddings**, mapping words into dense, continuous vector spaces where geometry mirrors semantic meaning.

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

$[0,$
 $0,$
 $1,$
 $1,$
 $0,$
 $\dots,$
 $0]$

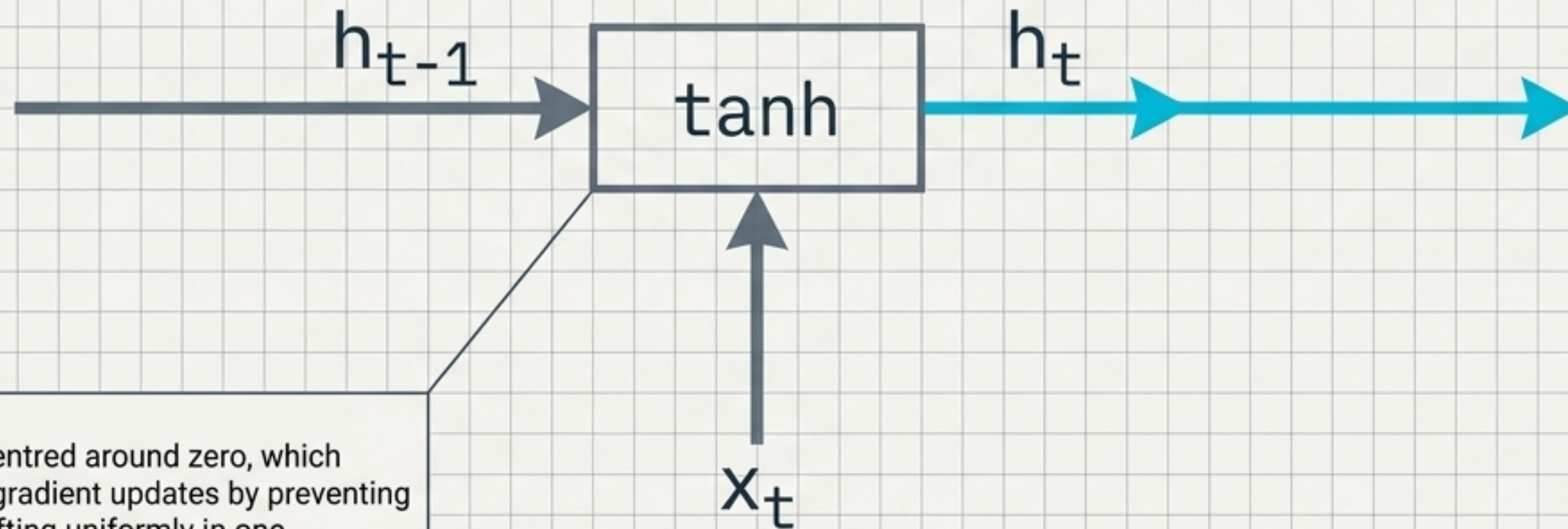
One-Hot Encoding.
Length =
Vocabulary Size.
Computationally
massive and
semantically blind.



The First Breakthrough: Recurrent Neural Networks

Hidden State:
$$h_t = \tanh(W_i * x_t + W_h * h_{t-1} + b)$$

Output:
$$\hat{y} = \text{softmax}(W_o * h_t + b_y)$$



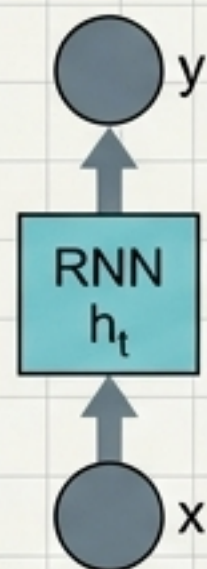
Why tanh?

The output is centred around zero, which helps stabilise gradient updates by preventing values from shifting uniformly in one direction.

The Topography of Sequence Modelling

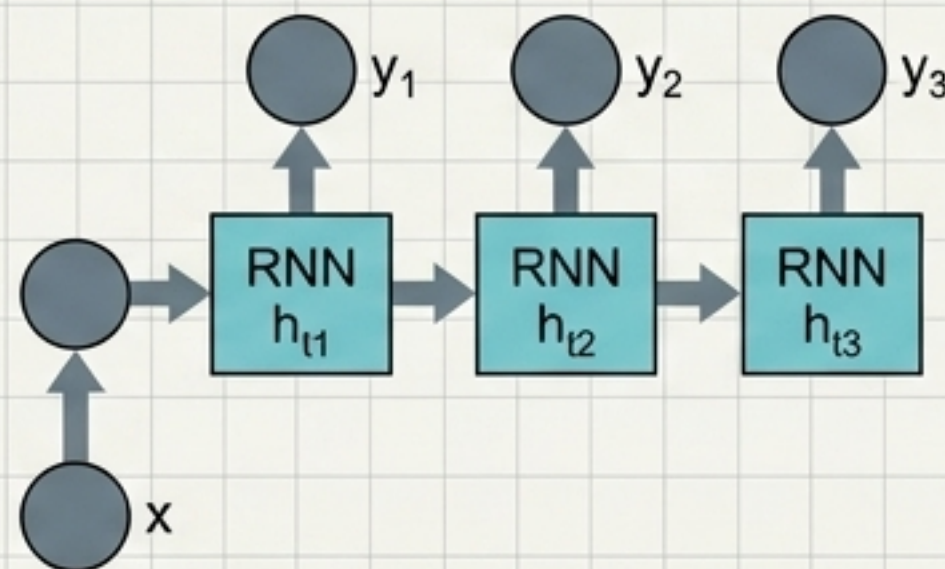
One-to-One

Standard static input (Image Classification)



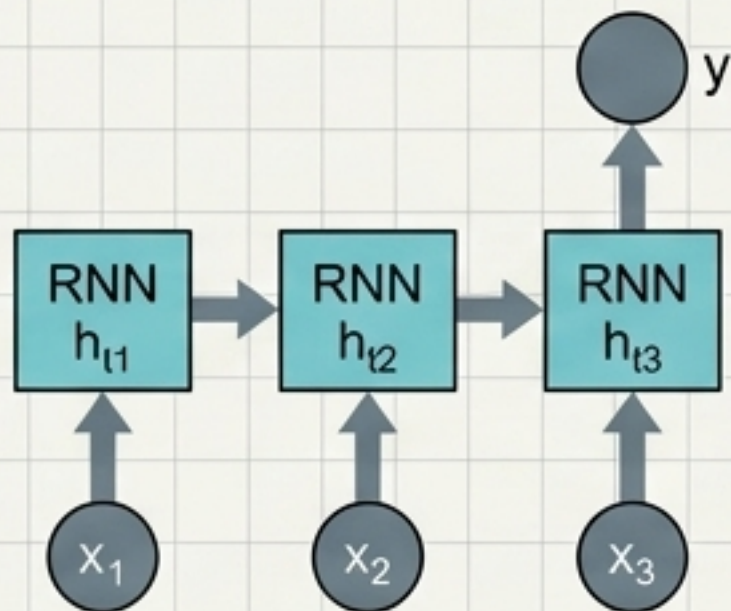
One-to-Many

Single input expands (Image Captioning)



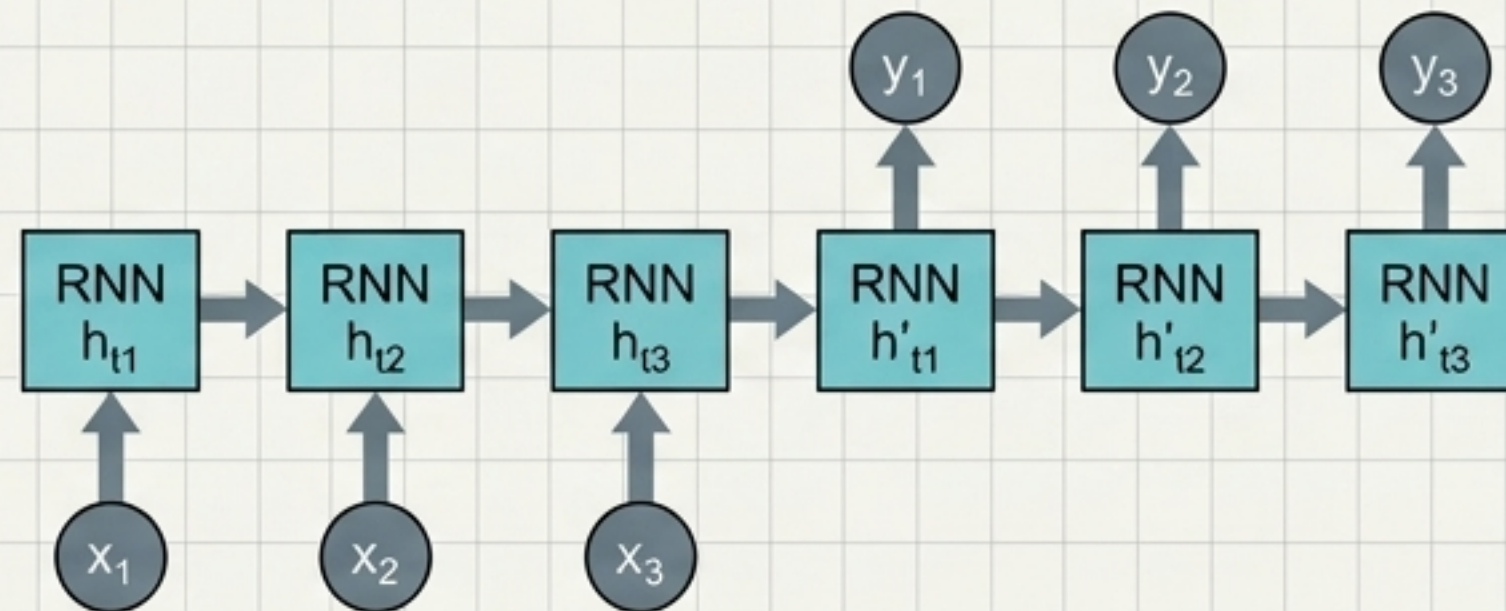
Many-to-One

Sequence collapses (Sentiment Analysis)



Many-to-Many (Encoder-Decoder)

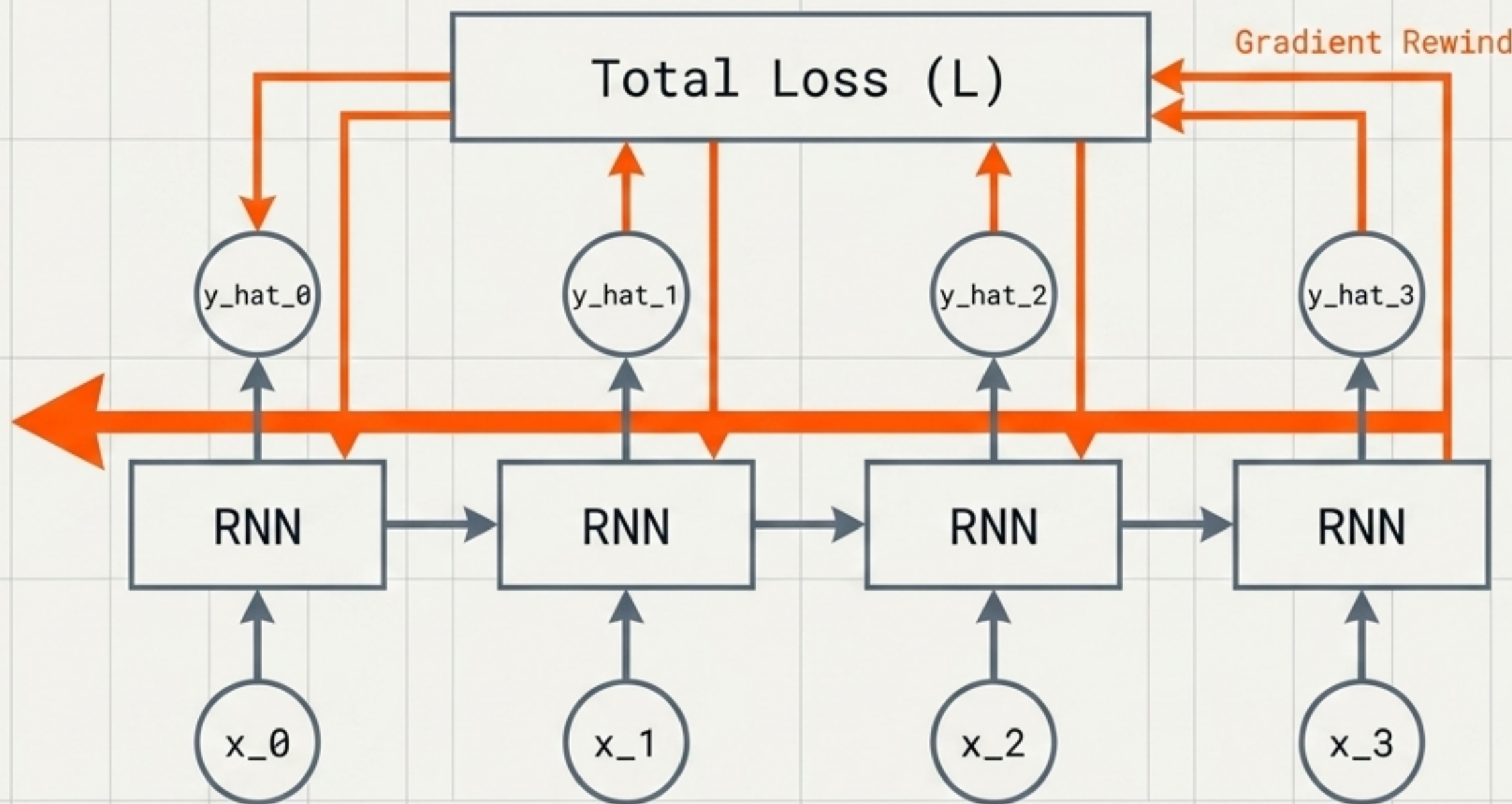
Asynchronous sequences (Machine Translation)



Training the Blueprint: Backpropagation Through Time

Specification

$$W_{\text{new}} = W_{\text{old}} - \alpha * \text{SUM}(dL/dW)$$



Padding & Bucketing: Variable-length sequences are padded with zeroes to match the longest sequence in the bucket, allowing parallel batch computation.

The Fatal Flaw: Vanishing Gradients

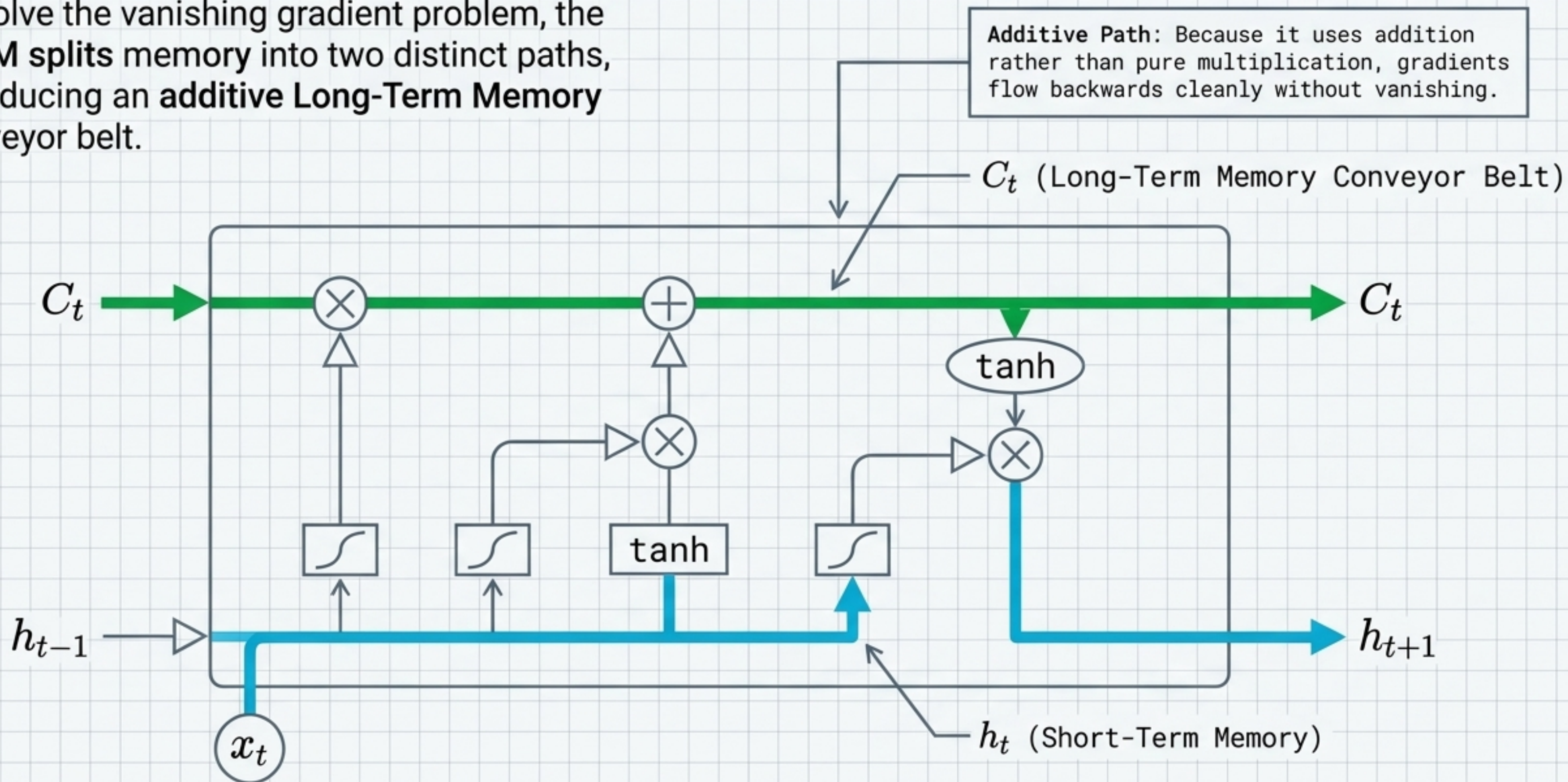
As BPTT rewinds through time, the gradients are repeatedly multiplied by fractions (tanh derivatives). The learning signal physically vanishes before reaching early context.



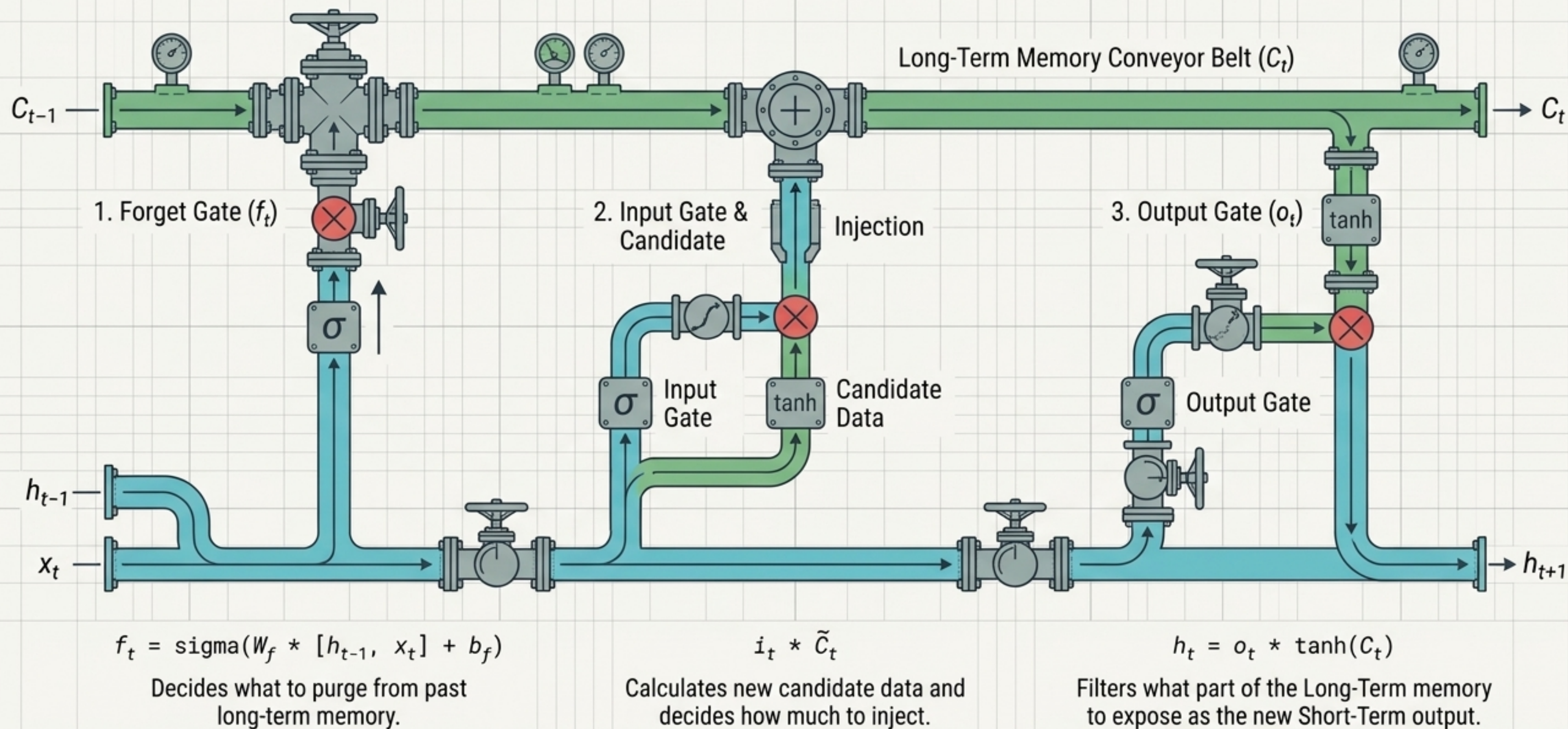
Exploding Gradients: The inverse structural problem where weight updates overflow to NaN. Solved mechanically via Gradient Clipping.

The Engineering Fix: Long Short-Term Memory (LSTM)

To solve the vanishing gradient problem, the **LSTM splits memory** into two distinct paths, introducing an **additive Long-Term Memory conveyor belt**.

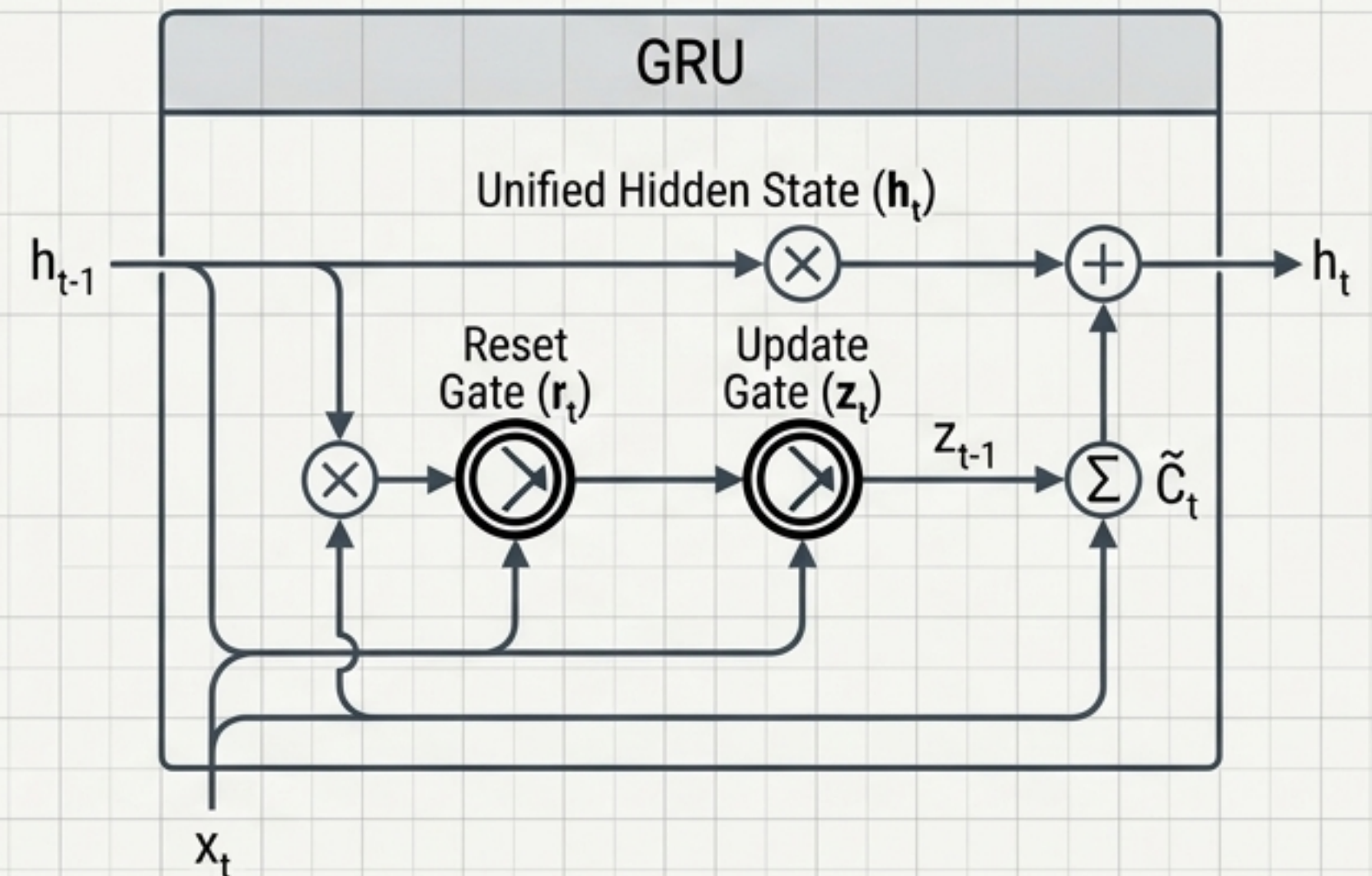
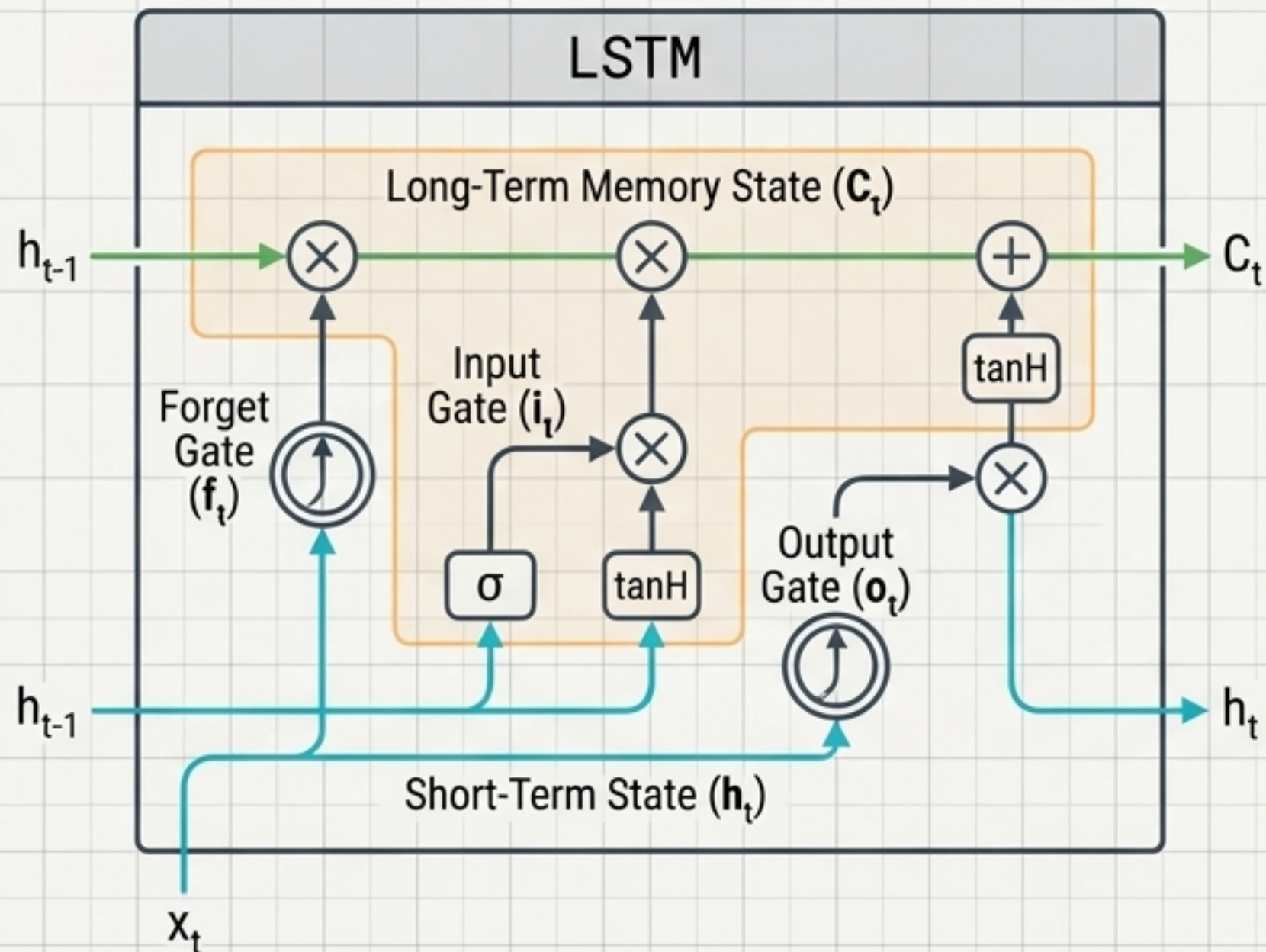


The LSTM Factory Line: Filtering the Stream



The Streamlined Engine: Gated Recurrent Unit (GRU)

The GRU simplifies the heavy LSTM architecture. It merges the dual cell state into a single vector and reduces the gate count from three to two, achieving similar performance with significantly faster training.



$$\text{Update: } z_t = \sigma(W_z * [h_{t-1}, x_t])$$

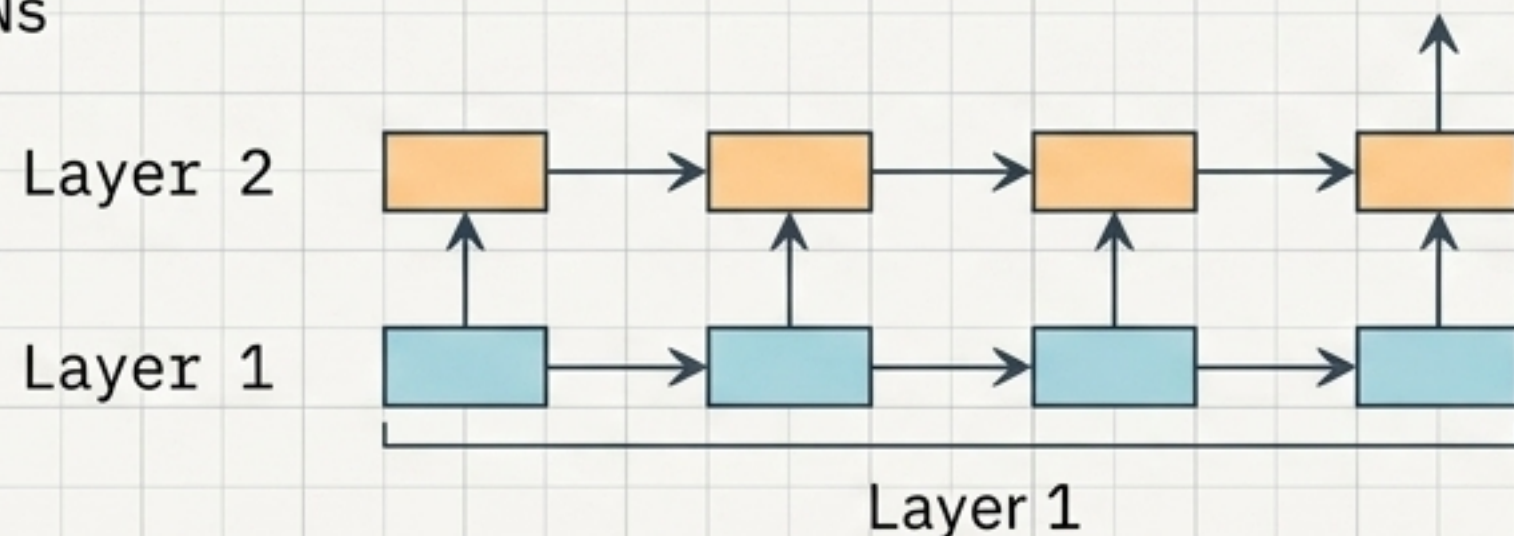
$$\text{New State: } h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{C}_t$$

The Cell Evolution Matrix

Architecture	Memory State	Gate Count	Long-Term Context	Computational Cost
Vanilla RNN	Single (h_t)	0	Poor (Vanishing Gradients)	Low / Fast Training
LSTM	Dual (h_t , c_t)	3 (Forget, Input, Output)	Excellent	High / Slow Training
GRU	Single (h_t)	2 (Reset, Update)	Good	Medium / Efficient

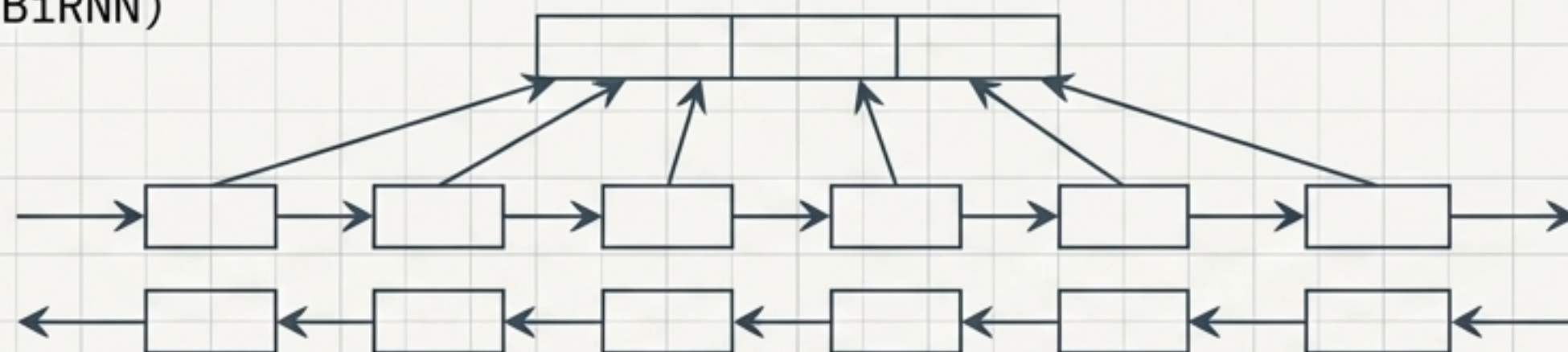
Architectural Expansions: Deep and Bidirectional Networks

Deep (Stacked) RNNs



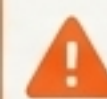
Increases representation power. Prone to overfitting; requires dropout. Ideal for high-complexity tasks like speech recognition.

Bidirectional RNN (BiRNN)



I love Amazon.

Is Amazon a company or a river?
BiRNNs consume the entire future context to decide.

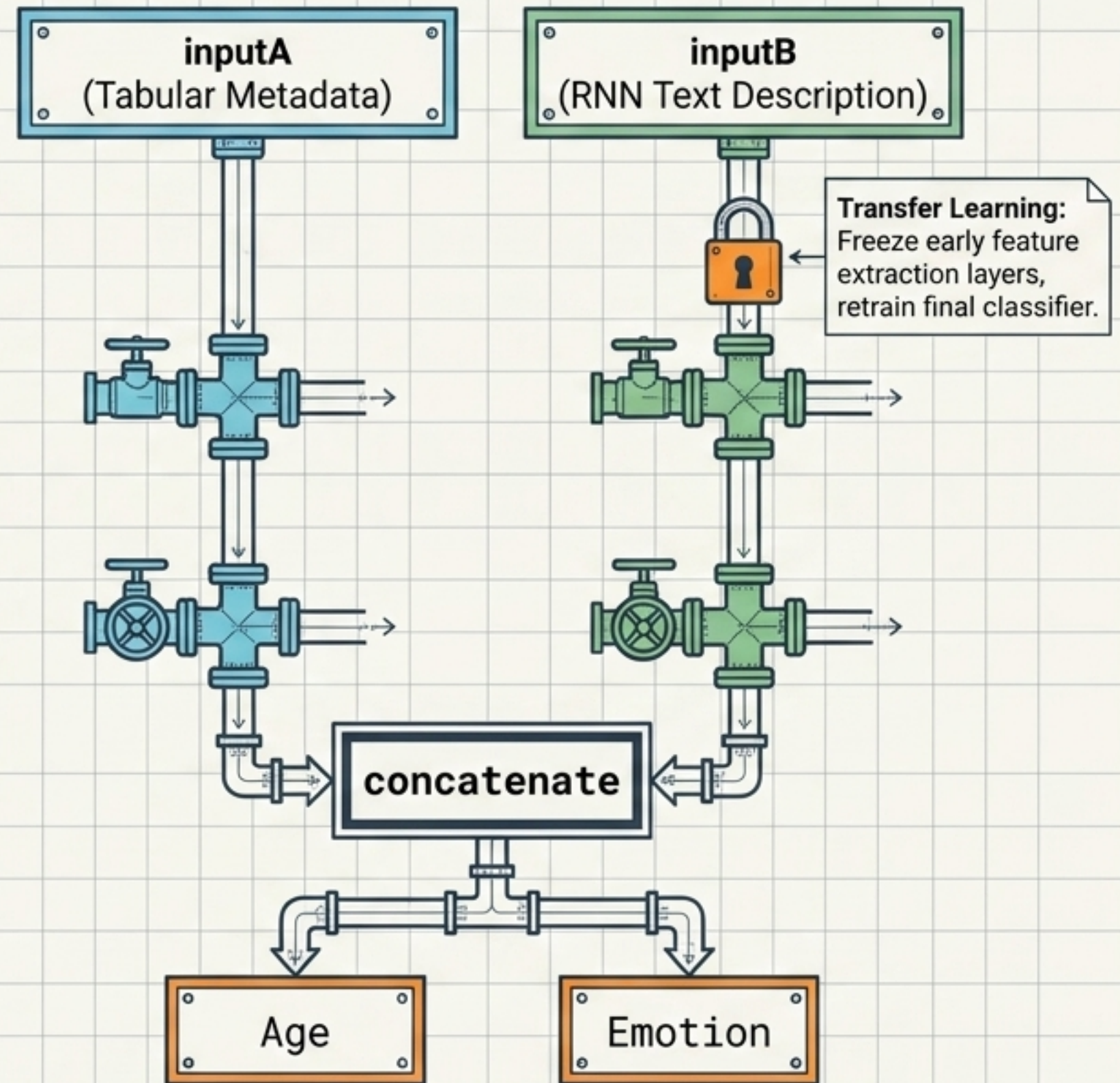


Cannot be used for real-time streaming.

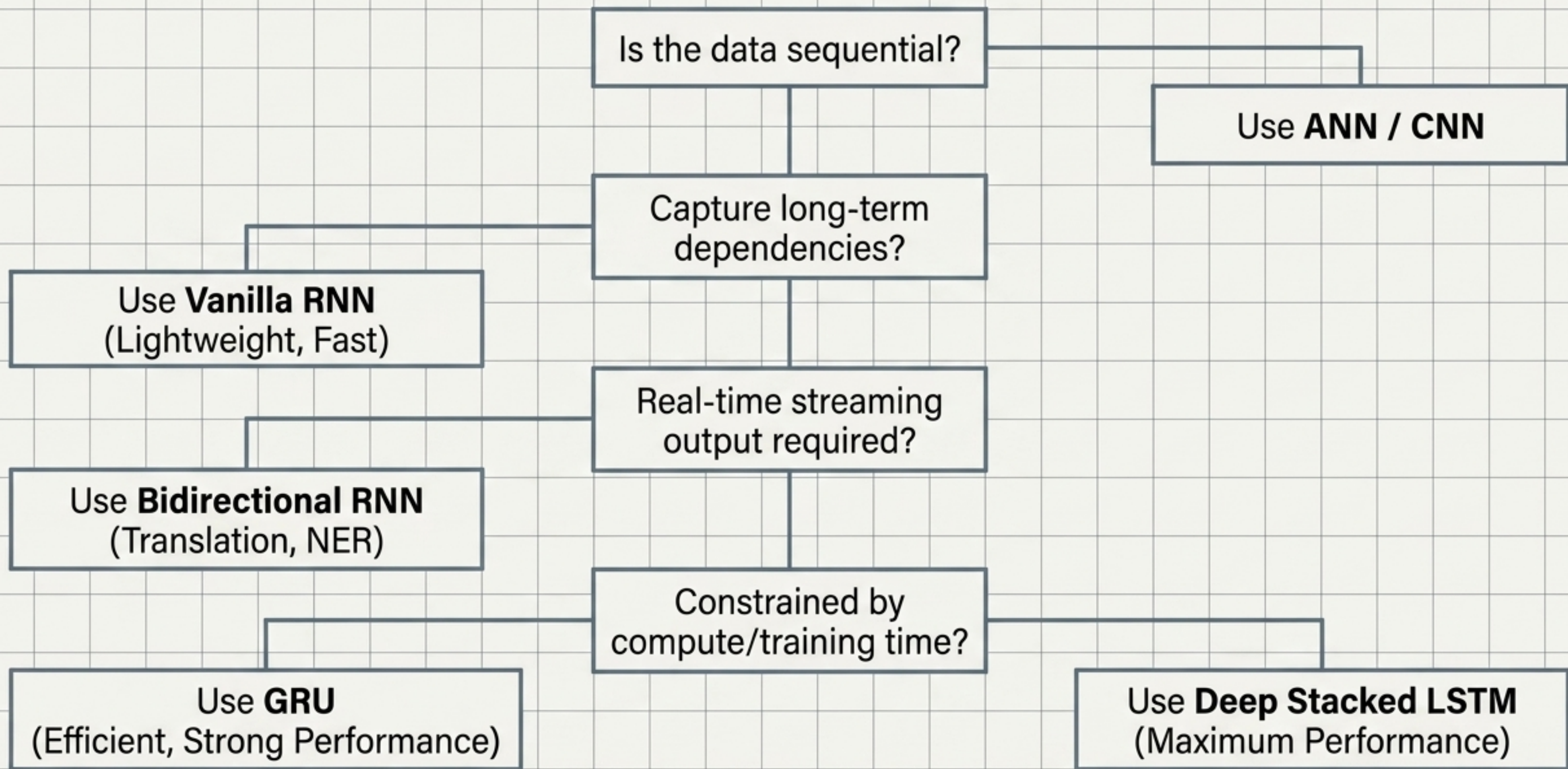
Building Complex Topologies: The Functional API

```
inputA = Input(shape=(32,)) # Tabular
inputB = Input(shape=(128,)) # RNN Text
[...]

combined = concatenate([x1, y2])
out1 = Dense(1, name='Age')(combined)
out2 = Dense(1, name='Emotion')(combined)
```



The Sequence Modeller's Toolkit



Architectural Reference & Key Takeaways

- **Standard NNs vs RNNs:** ANNs demand fixed inputs. RNNs dynamically unroll to process variable-length sequences, sharing weight parameters across all time steps.
- **Embeddings over One-Hot:** Learned dense vectors physically cluster semantically related terms, destroying the extreme dimensionality of sparse matrices.
- **The BPTT Mechanism:** Loss is computed per individual time step and summed. Gradients physically propagate backwards through time to update shared weights.
- **The Vanishing Gradient Limit:** Repeated multiplication of fractional tanh derivatives shrinks the learning signal to zero, destroying long-term memory in vanilla architectures.
- **The LSTM Innovation:** Introduces an additive Cell State (C_t) conveyor belt, rigorously regulated by three sigmoidal logic gates (Forget, Input, Output).
- **The GRU Optimisation:** Streamlines the architecture by merging hidden and cell states, utilising only two gates (Reset, Update) for vastly accelerated training.
- **Scaling Architecture:** Deep RNNs vertically stack layers to multiply representation power. Bidirectional RNNs process sequences forward and backward simultaneously for absolute context.